Notes

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$B \subseteq C \implies \mathscr{P}(B) \subseteq \mathscr{P}(C)$

Proof. By def $\mathscr{P}(B) = \{x \mid x \subseteq B\}$
$\forall x, x \in \mathscr{P}(B)$, Then, $x \subseteq B \subseteq C$
$\implies x, x \subseteq C$
$\implies x \in \mathscr{P}(C)$ By def $\mathscr{P}C = \{x \mid x \subseteq C\}$
Thus, $\mathscr{P}(B) \subseteq \mathscr{P}(C)$

□

A is a finite set,$|A| = n \implies |\mathscr{P}(A)| = 2^n$

Proof. Consider taking i elements from A to build a subset.
Then $i \in \{0, \ldots, n\}$, and the number of that kind of subsets is $\binom{n}{i}$.
(if $i = 0$, the subset is $\emptyset$; if $i = n$, the subset is A itself)
$\implies |\mathscr{P}(A)| = \sum_{i=0}^{n} \binom{n}{i}$
$\implies |\mathscr{P}(A)| = 2^n$

□

$x, y \in B \implies \{\{x\}, \{x, y\}\} \in \mathscr{P}(\mathscr{P}(B))$

Proof. $x, y \in B$
$\implies \{x\}, \{x, y\} \in \mathscr{P}(B)$
$\implies \{\{x\}, \{x, y\}\} \subseteq \mathscr{P}(B)$
Thus, by def $\{\{x\}, \{x, y\}\} \in \mathscr{P}(\mathscr{P}(B))$ □